

Pratham Das

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Full-Stack Developer with 2 years of experience building and shipping production applications in startup environments, with a strong focus on backend engineering. Experienced in designing REST APIs, database schemas, authentication, background processing, integrations, and performance-critical systems using Python (FastAPI, Django), Node.js, Express, and PostgreSQL. Strong frontend experience with React and Next.js, enabling end-to-end feature development across the stack.

PROJECTS

Raggy – Distributed RAG-Based Learning System

Github

- Designed and implemented an asynchronous ingestion architecture where user-uploaded content (YouTube, PDFs, audio, video, text) is queued, processed by Celery workers, converted to text (Whisper), and persisted in PostgreSQL and S3-compatible storage (MinIO).
- Built parallel worker pipelines for embedding generation and vector indexing (Qdrant) alongside automated summarization, enabling non-blocking processing and scalable concurrent workloads using Redis as broker.
- Engineered retrieval and QA services using FastAPI and LangChain, supporting contextual chat over user-provided data with authentication, rate limiting, structured logging, and protected LLM-backed endpoints.
- Integrated text-to-speech generation with object storage persistence, delivering both textual and audio responses while maintaining backend-driven orchestration of OpenAI-powered workflows.

EXPERIENCE

WhatBytes

April 2026 – Present

Backend Intern

- 6 production products owned across backend development and delivery, covering AI, hiring, communications, social, food delivery, and nonprofit platforms with Django, FastAPI, Node.js, PostgreSQL, Redis, Docker, and AWS.
- 20+ agent tools and 5+ secured APIs powering an agentic RAG platform built from scratch with LangGraph and LangChain, enabling fresh, permission-aware retrieval of candidate, job, recruiter, and application data across authenticated and unauthenticated workflows.
- 83% faster document generation, reducing processing time from 60+ seconds to 10 seconds and document size from 10 MB to 1 MB through caching, background processing, S3 lifecycle management, and image optimization.
- 20+ Microsoft Graph APIs and 30+ agent tools integrated into an AI communication platform with persistent thread-level memory, enabling responses that continuously adapt to conversation context, tone, formatting, and content.
- 10+ client requirement and feature discussions translated into technical architecture and implementation plans, with direct ownership of backend development, cross-functional coordination, and production delivery.

HashedTokens

Aug 2024 – April 2026

Founding Intern

- Built and shipped 10+ production-grade frontend applications, architected for scalability and fast iteration, enabling teams to ship features confidently without degrading performance over time.
- Consistently achieved 90+ Lighthouse scores across performance, SEO, and accessibility, improving load speed, search visibility, and user experience — directly supporting higher retention and conversion potential.
- Reduced client-side bundle size by 35% by restructuring rendering strategies and isolating interactivity, resulting in faster page loads on low-bandwidth devices and improved Core Web Vitals.
- Designed predictable state and data layers integrating REST, WebSockets, and Server-Sent Events, collaborating closely with backend engineers across different system architectures while maintaining 85%+ automated test coverage and Docker-based reproducible deployments.

EDUCATION

Kalyani Government Engineering College

West Bengal, India

Bachelor of Technology in Information Technology and Engineering -

Aug. 2022 – July 2026

TECHNICAL SKILLS

JavaScript, TypeScript, Python, Go, SQL, Bash, FastAPI, Django, Express.js, Gin, PostgreSQL, SQLite, Redis, MongoDB, Qdrant, S3/MinIO, LangChain, LangGraph, RAG, AWS, Docker, CI/CD, Git, Pytest